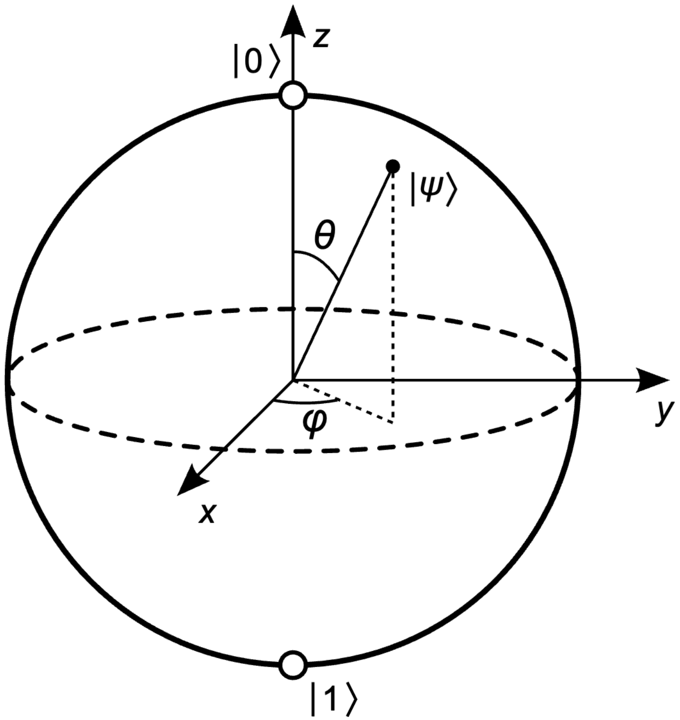
思考1

如果$d = m_1 \times m_2 \times \cdots \times m_k$，给定一个状态

$|\varphi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle + \cdots + \alpha_{d-1}|d-1\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{d-1} \end{pmatrix}$，如何判断$|\varphi\rangle$是系统 $\mathbb{C}^{m_1} \times \mathbb{C}^{m_2} \times \cdots \times \mathbb{C}^{m_k}$上的乘积态还是纠缠态?

思考2



$|\varphi\rangle = e^{i\gamma}(\cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle)$

I, X, Y, Z与 $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ 在Bloch sphere上的效果分别是什么？